

3.9 Blood Transfusion

▼ What determines an individual's blood group?

- The combination of **antigens on the surface of their red blood cells**.

▼ What two conditions can be caused by antibodies acting against red blood cells?

- **Haemolytic transfusion reactions (HTRs)**
- **Haemolytic disease of the foetus and newborn (HDFN)**

▼ What happens in haemolytic transfusion reactions (HTR)?

- The **antigens** on the transfused red blood cells are **recognised as foreign** which leads to haemolysis of these cells using **antibodies from the patient's plasma**

▼ What happens in haemolytic disease of the foetus and newborn (HDFN)?

- Antibodies from the mother cross the placenta and recognise the foetal RBCs as foreign which leads to haemolysis in the fetus
- This only occurs when the mother has a different blood group to the foetus.

▼ What are the two types of ABO antibody that can act against RBCs?

- Naturally occurring
- Acquired alloantibodies

▼ What are naturally occurring antibodies?

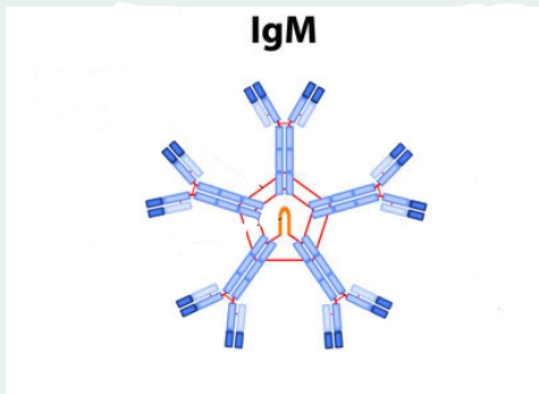
- These are **ABO antibodies (i.e anti-A, anti-B)** that exist naturally in humans and are used to combat antibodies on foreign RBC blood groups

▼ When are naturally occurring antibodies first produced?

- When the **host encounters them** in young age from **food and microorganisms**
- This is because these **sugars are very similar to those found on the antigens** of RBCs

▼ What type of antibody is usually a naturally occurring antibody?

- IgM



▼ What is a consequence of foreign blood group transfusion caused by naturally occurring antibodies?

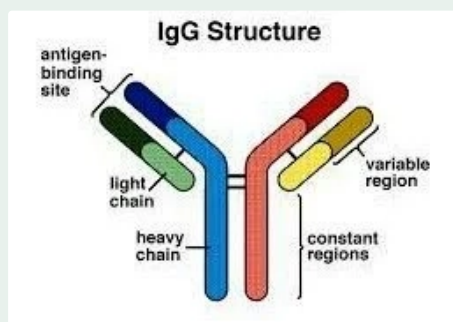
- **Haemolytic transfusion reaction (HTR)**
- This is because the antibodies recognise the foreign antigens and cause **agglutination** as well as **intravascular haemolysis** as they kill the foreign RBCs

▼ When are acquired alloantibodies initially produced?

- As a result of **active immunisation (alloimmunisation)** to 'non-self' **RBC antigens** following exposure to RBCs **from another individual**
- i.e.
 - When an **unfamiliar blood group is transfused** to a host
 - When an **unfamiliar fetal blood group crosses the placenta** into the mother's bloodstream.

▼ What type of antibody is usually an acquired alloantibody?

- IgG.

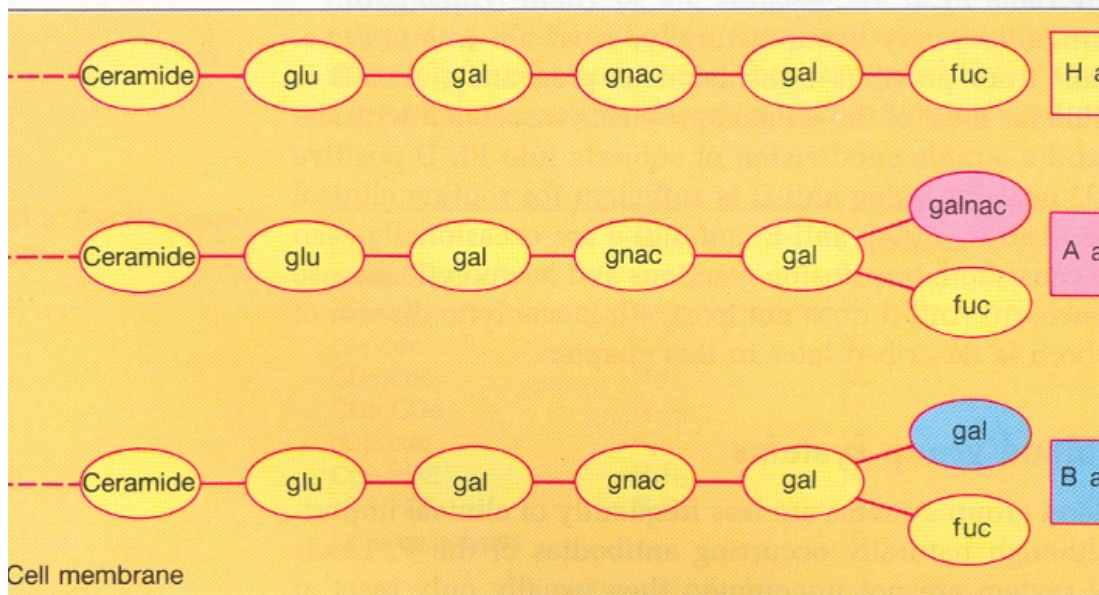


▼ Which antibody can cross the placenta to cause HDFNs?

- **IgG - acquired alloantibody**

- ▼ What is the difference between HTRs caused by naturally occurring antibodies and acquired alloantibodies?
 - HTRs caused by **naturally occurring antibodies** lead to **massive intravascular haemolysis**
 - This is because **IgM** antibodies are capable of **fully activating complement**
 - HTRs caused by **acquired alloantibodies** lead to mild intravascular haemolysis and **delayed HTRs**
- ▼ What are the four main blood groups within the ABO blood group system?
 - A, B, AB and O
- ▼ Which antigens are expressed by each of the cells in the blood groups?
 - A (A antigen)
 - B (B antigen)
 - AB (both A and B antigen)
 - O (neither antigen)
- ▼ What does the A gene code for?
 - The **A gene codes for an enzyme that adds N-acetyl galactosamine (GalNac)** to the **common H antigen** resulting in the A antigen
- ▼ What does the B gene code for?
 - The B gene codes for an enzyme that adds **galactose (Gal)** to the **common H antigen** resulting in the B antigen
- ▼ What does the O gene code for?
 - The **O gene produces an inactive enzyme** so the **H antigen remains unchanged** and **neither A nor B antigen** can be formed . The O gene is **recessive**.

- ▼ Image showing the differences between A, B and O blood groups.



- ▼ Which blood group is more prevalent in the British Caucasian population?
- **Groups A and O**
- ▼ Which blood group is more prevalent in the South East Asian population?
- **Group B**
- ▼ Which blood group is more prevalent in the African population?
- **Group O**
- ▼ Why may there be differences in the prevalence of blood groups in different areas?
- Different blood groups can give you different survival advantages in different areas
 - This is because selection pressures are different, and the advantageous areas varies with geographical distribution
- ▼ Recall two ways that blood groups can alter susceptibility to infection.
- **Receptors** on RBCs can **allow colonisation and invasion of pathogens**
 - **"False receptors"** on RBCs can **prevent pathogens binding to tissue**
- ▼ What is Landsteiner's law?
- It is a law that states that **whichever ABO antigens are lacking on a given person's RBCs, that person will always have the corresponding antibody.**
 - This is explained due to the exposure to these antigens on food and microorganisms at a young age.

- ▼ Recall two reasons why IgG antibodies that cross the placenta don't even end up causing HDFN most of the time.
 - **Foetal RBCs have poorly developed antigens which don't bind well to antibodies**
 - **ABO antigens are found on numerous other cells** (not just red cells) so any IgG ABO antibodies that have crossed the placenta can be 'mopped-up' by these cells without causing damage to fetal RBCs
 - Those antibodies that cross usually **disappear after a few months** as the **baby begins to develop its own antibodies**.
- ▼ Which blood group can be donated universally to all patients and why?
 - Group O
 - These contain neither A or B antigens and so there is no risk of causing acute HTRs.
- ▼ When is a platelet unit described as "high-titre negative"?
 - When there is a **low concentration of anti-A or anti-B antibodies in their plasma** and they are **unlikely to attack ABO antigens on RBCs**.
 - This is because platelets are usually administered with plasma (platelets are found in plasma)
 - On the contrary, **high-titre platelets** contain **lots of anti-A or anti-B antibodies** that are **likely to attack ABO antigens**
- ▼ Which should be used, high-titre or high-titre negative platelets and why?
 - **High-titre negative platelets**
 - This is because their plasma contains a **lower concentration of anti-AB antibodies than high-titre platelets** which means lower risk of HTR
 - NOTE: This only matters if the donor's plasma group does not match the recipient's plasma group. If the groups were matching, we'd have no concern as no matter how many antibodies are present in the plasma, they would not attack the host ABO antigens as the blood groups are the same.
- ▼ What is FFP?
 - **Fresh-frozen plasma.**

- ▼ What is the universal plasma donor group and why?
 - **Group AB plasma** as it **contains no antibodies**
 - This is because **Group AB individuals have both A and B antigens in their blood**, so it would make no sense for them to naturally produce antibodies against these antigens
- ▼ What is main type of antigen under the Rh system?
 - The **D antigen**.
- ▼ Is the gene for this antigen dominant or recessive?
 - **Dominant**
 - This means the **RhD positive phenotype** is coded for by a **dominant allele**.
- ▼ What is the name of antibodies that act against antigen D?
 - **Anti-D antibodies**
- ▼ What type of antibodies are anti-D antibodies?
 - **IgG**
- ▼ Do RhD negative individuals have naturally occurring anti-D antibodies?
 - No.
 - They only develop anti-D antibodies on exposure to anti-D antigen
- ▼ How long after exposure to antigen D does it take antibodies to appear?
 - 2-5 months.
- ▼ What is anti-D immunoglobulin and when is it used?
 - It is a treatment commonly used for pregnant RhD negative women to prevent the destruction of blood cells in their RhD positive foetus
- ▼ How does anti-D immunoglobulin work?
 - **Foetal RhD positive RBCs are destroyed** when a **mother's anti-D antibodies cross the placenta into the foetus**
 - **Anti-D immunoglobulin works by destroying any cells that have antigen D on them in the mother's circulation, preventing the mother from producing anti-D antibodies** and eliminating the risk of this cross-placental cell death.
 - **NOTE: This only works in some cases**, as the mother will need additional monitoring in other cases

▼ Which Rh blood phenotype can be transfused to anyone?

- **RhD negative**
- This is because RhD negative RBCs have no antigen D on them and so there is no risk of HTRs. Also, RhD positive individuals have no antibodies that work against antigen D so this is insignificant anyway

▼ Why do platelets for blood transfusion need to be of the same RhD group as the host, even though platelets do not express the RhD antigen?

- Some of the **components of the platelet units can contain RhD positive RBCs** that can lead to development of antibody (alloimmunisation) in **RhD negative patients**

▼ How does forward grouping work when testing for ABO antigens?

- The **A and B antibodies are added** to the **patient's red cells** once at a time and agglutination when an antibody is added indicates the **presence of a red blood cell antigen for that antibody**

▼ How does reverse grouping work when testing for ABO antigens?

- The **A and B antigens are added** to the **patient's serum antibodies** once at a time and agglutination when an antigen is added indicates the **presence of a antibody for that antigen**

▼ Figure showing how forward and reverse grouping works.

GROUP	FORWARD GROUP Patient red cells + reagents			REVERSE GROUP Patient's serum + reagents		
	Anti-A reagent	Anti-B reagent	Antigen(s) on RBC	Reagent A red cell	Reagent B red cell	Antibody in serum
O	No reaction	No reaction	No A or B antigen on RBC	Agglutination	Agglutination	Anti-A & Anti-B
A	Agglutination	No reaction	A antigen on RBC	No reaction	Agglutination	Anti-B
B	No reaction	Agglutination	B antigen on RBC	Agglutination	No reaction	Anti-A
AB	Agglutination	Agglutination	A and B antigen on RBC	No reaction	No reaction	No antibody

▼ How do we test for RhD groups?

- The anti-D reagent is added to the red cells.
- Agglutination will occur in RhD positive individuals but no agglutination occurs in RhD negative individuals.

BLOOD GROUP	Patient red cells + reagent anti-D antibodies	
	Anti-D reagent	Antigen on RBC
RhD positive	Agglutination	D antigen on RBC
RhD negative	No reaction	No D antigen on RBC

▼ What is the purpose of antibody screening and how does it work?

- Red blood cells that contain **all of the clinically relevant antigens are added to the patient's serum**
- Agglutination will occur when antibody interacts with antigen which **helps us know which acquired alloantibodies are present in the patient's serum** and transfuse based on this knowledge
- **Anti-human globulin** needs to be added as a **second reagent to make the agglutination visible** in this case.

▼ What is the purpose of a cross-match and how does it work?

- It is used in red-cell transfusions to assess a recipient's compatibility with the donor's red cells
- The **red cells are added to the patient's serum** and **agglutination is looked out for**
- **Agglutination** means there is the presence of an antibody which will reject the red cells and hence **incompatibility**
- **No agglutination** means there is no antibody which will reject the red cells and hence **compatibility**

▼ What is the difference between antibody screening and cross-match testing?

- Cross-match is different from antibody screening because whilst **antibody screening looks for the presence of all acquired alloantibodies**, but **cross-match only looks to see if the antibodies against the transfused RBC antigen is present**

- ▼ What is the difference between "group and screen" and "infection" testing?
 - **Group and screen testing** involves the **testing for antibodies** which may cause rejection
 - It also confirms the ABO and Rh blood groups of the patient
 - **Infection testing** is more directed at **blood-borne viruses and infections that may pose a threat to the recipient.**
- ▼ Image showing tests done on blood donors in the UK.

ROUTINE SCREENING TESTS (performed on all donations)	
	Anti-HIV 1+2 ab and PCR
HBs B (HBV)	HBV surface ag and PCR
HBs C (HCV)	Anti-HCV ab and PCR
HBs E (HEV)	PCR
Human T-lymphotrophic virus)	Anti-HTLV ab
Syphilis	Ab test (Treponema pallidum haemagglutination)
ADDITIONAL TESTS (only performed on some donations)	
Cytomegalovirus)	Anti-CMV ab
Chagas	Anti-T cruzii ab
Malaria	Anti-malarial ab

▼ ZONE BELOW

- ▼ What is apheresis?
 - A technique used to **separate blood components from each other**
- ▼ Describe how apheresis works.
 - **Blood from the donor passes through the machine which separates out the particular component required** (e.g. red cells, platelets, or plasma) and **returns the remainder of the blood to the donor.**
- ▼ Why are red cell transfusions usually required?
 - Red cell transfusions are required to **increase the haemoglobin concentration** in the recipient's blood
 - **1 unit of red cells** usually results in an **increase in the haemoglobin by 10g/L** in an average adult (weighing 70kg)

▼ How are red cells stored?

- **Red blood cells (RBCs) are stored in a fridge at 4°C.**

▼ What is the shelf-life of red blood cells?

- 35 days.

▼ Why are platelet transfusions usually required?

- Platelet transfusions are required for the treatment of bleeding or to **reduce the risk of bleeding in patients with low platelet counts** (thrombocytopenia) or **platelet dysfunction** (e.g. due to certain drugs such as antiplatelet medication)

▼ What are the two main methods of platelet donation?

- **Pooled platelets - platelets separated from 4 whole blood donations** are pooled together to make up 1 unit of platelets
 - i.e the primary aim of the donation was not platelet donation
- **Apheresis - a single donor donates 1 unit of platelets through apheresis**
 - i.e the primary aim of the donation was platelet donation

▼ How are platelets stored?

- Platelets are **stored at room temperature (22°C)**
- Platelets require constant agitation to ensure the **platelets are continuously oxygenated** within their **gas permeable containers**.

▼ What is the shelf-life of platelets?

- 7 days.

▼ Why are FFP transfusions usually required?

- **FFP contains all the coagulation factors** so transfusions of FFP are usually conducted to **reduce the risk of bleeding**.
- It can be used in **patients with coagulopathies** such as **increased PT/APTT**

▼ In response to what is FFP usually used?

- **Acquired coagulopathy**
 - i.e disseminated intravascular coagulation (DIC) which is abnormal blood clotting **caused by injury**
- **Inherited coagulopathy**
 - i.e abnormal clotting due to a factor deficiency such as Factor V deficiency

▼ What is the shelf-life of FFP/cryoprecipitate?

- 3 years.

▼ How is FFP/cryoprecipitate stored?

- **FFP must be stored at 4°C**
- However, it has a **shelf-life of 3 years at -25°C**
 - Once **thawed**, FFP **must be used within 24 hours**

▼ Why are cryoprecipitate transfusions usually required?

- Cryoprecipitate contains **fibrinogen, Factor VIII, von Willebrand factor (VWF) and Factor XIII.**
- Cryoprecipitate transfusions can hence be used to **treat or reduce the risk of bleeding**

▼ Which coagulation factors are found in cryoprecipitate?

- **Fibrinogen**
- **Factor VIII**
- **von Willebrand factor (VWF)**
- **Factor XIII.**

▼ In response to what is cryoprecipitate usually used?

- Low **fibrinogen**
- **Fibrinogen** dysfunction.

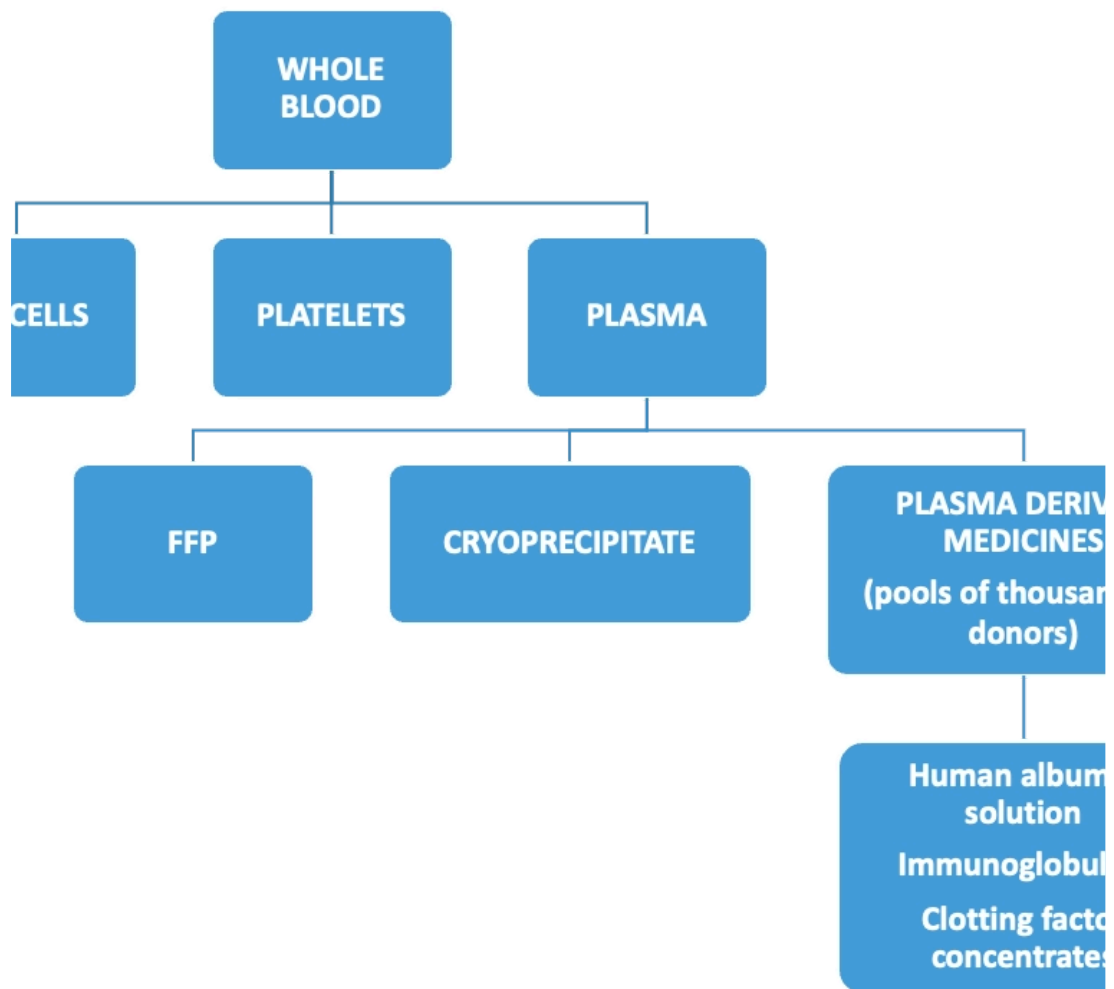
▼ What are the two main methods of cryoprecipitate donation?

- **Pooled donations** - 5 whole blood units put together
- **Single donation**

▼ How is cryoprecipitate made?

- Cryoprecipitate is made by **slow thawing of FFP overnight until there is a precipitate.**
- This precipitate is then **re-suspended in a small volume of plasma** and **frozen to maintain the activity of the clotting factors.**

- ▼ Image showing the use of all the blood component transfusions.



- ▼ What is the use of human albumin solution (HAS)?

- To increase plasma/albumin levels of the recipient.
- It can be done
 - To **restore plasma volume** in patients with plasma volume loss e.g. due to burns or trauma
 - To **replace plasma in plasma exchange** e.g in the treatment of autoimmune disorders
 - To **initiate diuresis** in **patient with low albumin**

- ▼ What is the use of immunoglobulin solution?

- This **contains antibodies to viruses** that need to be fought against in the recipient

▼ What are the two types of immunoglobulin solution?

- **Normal immunoglobulin**
 - This contains **antibodies to viruses that are common in the population.**
- **Specific immunoglobulins**
 - This is made from selected donors with **high antibody levels to the target of treatment**

▼ What are clotting factor concentrates?

- These are concentrates used to **replenish the level of a single or multiple clotting factors**
 - i.e Fibrinogen concentrates, single factor concentrates

▼ What do Prothrombin complex concentrates (PCCs) contain?

- **Factors II, VII, IX and X.**

▼ What can Prothrombin complex concentrates (PCCs) be used to combat?

- **Prothrombin complex concentrates (PCCs)** can be used to **reduce the effect of anticoagulants** such as warfarin.
- This is because **it contains all the liver-synthesised clotting factors that rely on Vitamin K** for carboxylation, and so it can **counter the effects of warfarin as a vitamin K antagonist**

▼ What is an alternative to cryoprecipitates that can be used to replace fibrinogen in patients?

- **Fibrinogen concentrates** provide an alternative to cryoprecipitate for replacing fibrinogen

▼ What are FFP and cryoprecipitate frozen?